



Problem-Solution Fit

Date	15 March 2026
Team ID	N/A
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Students • Teachers / Educators • Self-learners • Educational Institutions	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited time for evaluation. • Different accents and speaking styles. • Budget constraints. • Basic devices and internet availability.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros: • Manual teacher evaluation gives detailed feedback. • MCQs are easy to conduct. Cons: • Time-consuming. • Subjective evaluation. • Cannot accurately assess spoken conceptual understanding.
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**2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]**

- Difficult to evaluate conceptual understanding from spoken explanations.
- Manual assessment is slow and subjective.
- No instant feedback after speaking.
- Happens frequently during classes, exams, and practice sessions.

**9. PROBLEM ROOT /
CAUSE [RC]**

- No AI-based tool for spoken concept evaluation.
- Manual evaluation is inconsistent.
- Difficult to measure conceptual understanding objectively.
- Lack of automated feedback and scoring.

**7. BEHAVIOR
+ ITS INTENSITY [BE]**

- Students explain concepts verbally (High).
- Teachers assess multiple students (High).
- Students seek immediate feedback (High).
- Regular practice before exams (Medium).

**3. TRIGGERS TO ACT
[TR]**

- Upcoming exams or viva.
- Need for quick concept assessment.
- Teacher wants faster evaluation.
- Student wants self-assessment and improvement.

**10. YOUR SOLUTION
[SL]**

- Voice Based Concept Understanding Analyser
- Converts speech to text.
- Evaluates conceptual understanding using AI/NLP.
- Measures speech quality and concept similarity.
- Generates score, feedback, and improvement suggestions.
- Produces downloadable reports for students and educators.

**8. CHANNELS OF
BEHAVIOR [CH]**

ONLINE

- Streamlit Web Application
- College LMS
- Educational Websites

**4. EMOTIONS
BEFORE / AFTER [EM]
Before:**

- Confused
- Anxious
- Unsure of understanding

After:

- Confident
- Motivated
- Better understanding of concepts

OFFLINE

- Classrooms
- Labs
- Viva examinations